SIGHT.SERIALIZATION · SPECIFICATION

SBDL Text Format Specification

FORMAT

Sight Bundle, text encoding (`.sbdt`) and schema language (`.sgs`)

FORMAT VERSION

1

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SBDL Binary Format Specification

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1. Introduction

This document defines two textual languages sharing the same lexical structure:

- the **schema language** (`.sgs` files), used to declare schemas (§4);
- the **text data format** (`.sbdt` files), the human-readable equivalent of the binary format (§5–§8).

The data model, the schema model, schema evolution and resource resolution are defined by the binary specification (§3, §4, §9 and §10 of that document) and are not repeated here. A text file and a binary file holding the same data are **equivalent**: conversion in either direction is lossless, except for encoding details that have no textual meaning (§9).

1.1 Conventions

The key words **MUST**, **MUST NOT**, **SHOULD**, **SHOULD NOT** and **MAY** are to be interpreted as described in RFC 2119.

A document violating a **MUST** is **invalid**; a reader **MUST** reject it with an error reporting at least the line and column of the problem, and **MUST NOT** return a partial result.

Grammar rules use EBNF: `a b` sequence, `a | b` alternative, `[a]` optional, `{a}` zero or more, `"x"` literal text. The complete grammar is given in Appendix A.

2. Lexical structure

2.1 Encoding

- Files are UTF-8, **without** byte order mark. A BOM is an error.
- Line breaks are `LF` or `CR LF`. A lone `CR` is an error.
- Whitespace is space (`U+0020`) and tab (`U+0009`). Line breaks are significant only as separators (§2.6).

2.2 Comments

- `//` up to the end of the line.
- `/* … */`, possibly over several lines, not nested.

Comments are equivalent to whitespace (a block comment containing a line break counts as a line break).

2.3 Identifiers and keywords

```
Ident = Letter { Letter | Digit }          Letter = "A".."Z" | "a".."z" | "_"
QName = Ident { "." Ident }
```

Reserved words, which can never be used as an identifier value (enum value, alias, anchor): `null` , `true` , `false` , `inf` , `nan` .

Type keywords, which cannot be used as user type names: `bool`, `u8`, `u16`, `u32`, `u64`, `s8`, `s16`, `s32`, `s64`, `f32`, `f64`, `string`, `list`, `map`, `ref`, `extref`, `any`.

Other words (`schema`, `import`, `enum`, `struct`, `abstract`) are keywords only where the grammar expects them; they remain valid field names.

2.4 Literals

Integers

```
Integer   = [ "-" ] ( Decimal | Hex | Binary ) [ IntSuffix ]
Decimal   = Digit { [ "_" ] Digit }
Hex       = "0x" HexDigit { [ "_" ] HexDigit }
Binary    = "0b" BinDigit { [ "_" ] BinDigit }
IntSuffix = "u8" | "u16" | "u32" | "u64" | "s8" | "s16" | "s32" | "s64"
```

- `_` is only allowed between two digits. Leading zeros are allowed in decimal (`007`).
- Hex digits are case-insensitive. Only integer suffixes are allowed on hex and binary literals (`u` and `s` are not hex digits, so `0xFFu8` is unambiguous).

Floats

```
Float      = [ "-" ] ( Decimal "." Decimal [ Exponent ] | Decimal Exponent | "inf" ) [
FloatSuffix ]
           | "nan" [ FloatSuffix ]
Exponent   = ( "e" | "E" ) [ "+" | "-" ] Decimal
FloatSuffix = "f32" | "f64"
```

- Digits are required on both sides of `.` (`.5` and `5.` are invalid).
- A decimal integer literal **MAY** carry a float suffix (`2f32`).
- Parsing **MUST** be correctly rounded (round-half-to-even) to the target precision.

Strings

- Single-line: `"…"`. Escapes: `\"`, `\\`, `\n`, `\r`, `\t`, `\0`, `\u{X}` (1 to 6 hex digits, a Unicode scalar value, surrogates excluded). Any other escape is an error. Raw characters `U+0000`–`U+001F` (except tab) and line breaks are not allowed inside.
- Multi-line: `"""` immediately followed by a line break; the content ends at a line containing only whitespace followed by `"""`. The whitespace before the closing `"""` is the **indentation**: it is removed from the start of every content line, and every non-empty content line **MUST** start with it. The line break before the closing line is not part of the value. Escapes are processed as in single-line strings.

```
description: """
    First line
      indented line
```

```
    Last line
    """
```

Value: `"First line\n indented line\nLast line"`.

Base64

`b64"…"`: standard alphabet (RFC 4648 §4) with mandatory padding. Whitespace and line breaks inside the quotes are ignored. Only valid where a `list<u8>` is expected.

2.5 Punctuation

`{ } [ ] < > ( ) : , = ? & * # @`

2.6 Separators

Items of a list, map or struct body, enum values and directives are separated by a comma **or** at least one line break. A trailing comma before the closing bracket is allowed. Two items on the same line without a comma are an error.

```
[1, 2, 3]          // valid
[1 2 3]            // invalid
[
  1
  2,
  3,
]                  // valid
```

Two additional line-sensitive rules:

- A **type prefix** (§6.4) and the `{` that follows it **MUST** be on the same line.
- A **directive** (§5.1) starts at the beginning of a line (after optional whitespace) and ends at the end of that line, except for a `{ … }` block, which may span several lines.

3. Type expressions

Shared by the schema language and the data format:

```
Type      = BaseType [ "?" ]
BaseType  = "bool" | "u8" | "u16" | "u32" | "u64" | "s8" | "s16" | "s32" | "s64"
          | "f32" | "f64" | "string" | "any"
          | "list" "<" Type ">"
          | "map" "<" Type "," Type ">"
          | "ref" "<" QName ">"
          | "extref" "<" QName ">"
          | QName                              // enum or struct
```

Constraints of the data model apply (binary spec §3.1): no `any?`, map keys restricted, `ref` / `extref` targets are structs.

Type name resolution. An unqualified name is looked up in the current schema, then in its direct imports. If it is found in more than one place, it is ambiguous and **MUST** be qualified (`assets.Texture`). A qualified name is `<schema name>.<type name>`.

4. Schema language (`.sgs`)

4.1 Grammar

```
SchemaFile  = "schema" QName Decimal { Import } { TypeDecl }
Import      = "import" QName Decimal [ Hex ]
TypeDecl    = EnumDecl | StructDecl

EnumDecl    = "enum" Ident ":" IntType "{" [ EnumValue { Sep EnumValue } [ "," ] ] "}"
EnumValue   = Ident [ "=" Integer ]

StructDecl  = [ "abstract" ] "struct" Ident [ ":" QName ]
              "{" [ FieldDecl { Sep FieldDecl } [ "," ] ] "}"
FieldDecl   = Ident ":" Type [ "=" Literal ] { Annotation }
Annotation  = "@fixed" | "@alias" "(" Ident { "," Ident } ")"
```

`Sep` is a separator per §2.6. `Literal` is a scalar value literal (§6.2): boolean, number, string or enum value.

4.2 Semantics

Mapping to `core.Schema` (binary spec §4.5):

SOURCE	`CORE.SCHEMA`
`schema game 3`	`name = "game"`, `version = 3`
`import assets 2`	`SchemaRef { name: "assets", version: 2, hash }` — hash computed from the resolved import; if a hex hash is written, it **MUST** match
type declarations	`types`, in declaration order
`enum E : u8 { A, B = 5, C }`	`EnumDef`; a value without `=` is the previous value + 1, the first one defaults to 0 → `A = 0, B = 5, C = 6`
`abstract struct B : A { … }`	`StructDef { base: index of A, isAbstract: true, … }`
`x: u16 = 1`	`Field.default = 1`
`x: u64 @fixed`	`TypeRef.flags` bit 1

SOURCE	`CORE.SCHEMA`
`x: T @alias(old, older)`	`Field.aliases = ["old", "older"]`

Rules:

- All model rules apply (binary spec §4): unique names, acyclic inheritance, defaults restricted to scalar, string and enum types.
- A field **MUST NOT** be both nullable and defaulted (`T? = v` is invalid): `null` is the implicit default of a nullable field.
- `@fixed` is only valid on `u16`, `u32`, `u64`, `s16`, `s32`, `s64`, possibly nullable. Each annotation appears at most once per field.
- Types may be referenced before their declaration.

4.3 Example

```
schema game 3
import assets 2

enum Class : u8 { Warrior, Mage }

struct Item {
  name:  string
  count: u16 = 1
}

abstract struct Character {
  name:  string
  level: u8 = 1
}

struct Warrior : Character { rage: u16 }
struct Mage : Character { mana: u32 }

struct Player {
  party:      list<Character>
  inventory:  list<Item>
  stats:      map<string, f32>
  leader:     ref<Character>?
  icon:       extref<Texture>          // resolved in assets
  hash:       u64 @fixed
  experience: s32 @alias(xp)
  extra:      any?
}
```

5. Data document (`.sbdt`)

5.1 Structure

```
Document  = FormatDirective { Directive } RootValue { ExportDirective }
```

DIRECTIVE	OCCURRENCES	BINARY EQUIVALENT
`@sbdl 1`	exactly once, **first line** of the file	magic + format version
`@id` String	at most once	header `id` (default `""`)
`@version` Decimal	at most once	header `version` (default 0)
`@meta` (Ident | String) `=` String	any; keys unique	metadata
`@schema` QName Decimal [Hex]	at most once	schema mode Reference (name, version, optional hash)
`@schema` `{` SchemaFile `}`	one or more	schema mode Inline; the block contains a schema in the schema language
`@dep` Ident `=` (QName | String) Decimal	any; aliases and ids unique	dependency list, in order
`@export` (Ident | String) `=` Value	any, **after** the root	exports, in order

- `@sbdl` **MUST** be the very first characters of the file (no comment or blank line before it). The format version **MUST** be `1`.
- Directives other than `@export` **MUST** appear before the root value, in any order. A reference `@schema` and inline `@schema` blocks **MUST NOT** be mixed. No `@schema` at all means schema mode None.
- With inline schemas, the first block is the file schema. Imports not provided as an inline block are resolved externally (binary spec §10).
- With a reference `@schema`, the hash is verified when present. Writers **SHOULD** omit it.
- A `@dep` **alias** is local to the file and is only used by `extref` values (§7.2). The dependency **id** is what is stored in the binary header.

Search path indexing (binary spec §10.3) reads `.sbdt` files up to the first token that is not a header directive.

5.2 Root value

The root value is a struct value with a **mandatory type prefix** (`Player { … }`), since schemas do not declare a root type. The type **MUST** be a non-abstract struct.

5.3 Exports

`@export name = value`, where `value` is either `*anchor` (§7.1) or a struct value with a mandatory type prefix, optionally anchored. Export names **MUST** be unique.

6. Values

6.1 Grammar

```
Value      = "null"
           | Scalar
           | List
           | Braced
           | Anchor Braced
           | Reference
           | ExtRef
           | TypedValue                          // only where the type is `any`, §6.5

Scalar     = "true" | "false" | Integer | Float | String | MultiString | Base64 | EnumValue
EnumValue  = QName                               // not followed by "{" on the same line
List       = "[" [ Value { Sep Value } [ "," ] ] "]"
Braced     = [ QName ] "{" [ Entry { Sep Entry } [ "," ] ] "}"
Entry      = Key ":" Value
Key        = Ident | String | Integer | "true" | "false"
Anchor     = "&" ( Ident | Decimal )
Reference  = "*" ( Ident | Decimal )
ExtRef     = "<" Ident [ "#" ( Ident | String ) ] ">"
```

The grammar is unambiguous without a schema: a document can always be parsed into a syntax tree. The schema is required to **type** the tree: it decides whether `12` is a `u8` or an `f32`, whether `{ … }` is a struct or a map, and whether `Mage` is an enum value.

6.2 Typing rules

EXPECTED TYPE	ACCEPTED SYNTAX
`bool`	`true`, `false`
integer types	Integer literal within the range of the type; a suffix, if any, **MUST** equal the type
`f32`, `f64`	Float or decimal Integer literal; a suffix, if any, **MUST** equal the type
`string`	String or MultiString
enum `E`	value name of `E` (`Mage`) or qualified with the enum name (`Class.Mage`, or `game.Class.Mage`)

EXPECTED TYPE	ACCEPTED SYNTAX
`list<u8>`	List of integers or Base64
`list<T>`	List
`map<K, V>`	Braced **without** type prefix; keys typed as `K`: String for `string`, Integer for integers, `true` / `false` for `bool`, Ident for enums
struct `T`	Braced with Ident keys (field names); type prefix rules in §6.4
`T?`	`null` or any syntax accepted for `T`
`ref<T>`	§7.1
`extref<T>`	§7.2
`any`	§6.5

A literal out of range for its type, or a syntax not accepted for the expected type, is an error. Integer literals are exact: `1.0` is not accepted for an integer type.

6.3 Struct fields

- Fields **MAY** appear in any order. A field **MUST NOT** appear twice.
- A field name that is not a field of the concrete type (including inherited fields) is an error. Aliases are **not** accepted in text: a text document is always written with the schema it references.
- A missing field takes `null` if nullable, its default if it has one, and is an error otherwise.

6.4 Type prefixes

A struct value **MAY** be preceded by its type name, on the same line as `{`:

- **mandatory** for the root value, exports, struct values in `any`, and when the concrete type differs from the declared type (`Warrior { … }` in a `list<Character>`);
- **optional** otherwise; if present, it **MUST** be the declared type or a type deriving from it. If the declared type is abstract, a prefix is always mandatory.

6.5 Values of type `any`

The value carries its own type:

SYNTAX	TYPE
`null`	Null
`true`, `false`	`bool`
Integer without suffix / with suffix	`s64` / the suffix type

SYNTAX	TYPE
Float without suffix / with suffix	`f64` / the suffix type
String, MultiString	`string`
Base64	`list<u8>`
`E.Value` (qualified enum value)	enum `E`
`T { … }` (mandatory prefix)	struct `T`
`[ … ]`	`list<any>`
`{ … }` with String keys	`map<string, any>`
`*name`, or `&name T { … }`	`ref<T>`, `T` being the concrete type of the target
TypedValue: `Type Value`	the explicit `Type`

```
TypedValue = Type Value      // e.g. list<u8> [1, 2], map<s32, f32> { 1: 0.5 },
extref<assets.Texture> <tex>
```

An explicit type is required for any other type, e.g. `list<s32> [1, 2]`, `map<Class, u8> { Mage: 1 }` or `extref<Texture> <tex>`. An unqualified enum value or an unprefixed struct is an error in `any`.

7. References

7.1 Internal references

- `&name` (anchor) is placed before an object at a `ref<T>` position, an export, or an `any` value; the object becomes referenceable.
- `*name` refers to an anchored object. The concrete type of the target **MUST** be `T` or derive from `T`.
- Anchor names are identifiers or decimal numbers, local to the document, and **MUST** be unique.
- References **MAY** precede their anchor (forward references); every reference **MUST** resolve once the document is parsed.
- An anchor on a value that is not at a `ref<T>` position, an export or an `any` value is an error.
- In `any`, an anchored struct (`&a T { … }`) is a `ref<T>`; an unanchored struct (`T { … }`) is a struct value.

The position of the anchor in the text has no meaning: it does not need to match the position where a binary writer emits the object inline.

7.2 External references

- `<alias>`: root value of the dependency declared with `@dep alias = …`.
- `<alias#name>` or `<alias#"any name">`: export `name` of that dependency.

- An undeclared alias is an error.

8. Canonical output

Writers **MUST** produce the canonical form defined here, so that equal data gives identical text (stable diffs).

Layout

- `LF` line breaks, final line break, no trailing whitespace, indentation of 2 spaces per level (configurable).
- Directives in this order: `@sbdl 1`, `@id` (if non-empty), `@version` (if not 0), `@meta` (in order), `@schema`, `@dep` (in order); then one blank line, the root value, and, if any, one blank line followed by the `@export` directives.
- A list, map or struct is written on a **single line** if its whole representation fits within the maximum width (default 80 columns, measured from the start of the line) and contains no multi-line string: `[1, 2, 3]`, `{ name: "Potion", count: 3 }` (one space inside braces, none inside brackets). Otherwise it is written with one item per line, without commas.
- Empty containers: `[]` and `{}`.

Values

- Struct fields in field sequence order (binary spec §4.2); fields equal to their default and null nullable fields are omitted (unless configured otherwise).
- Type prefixes only where mandatory (§6.4).
- Map entries in their order; enum values unqualified unless in `any`.
- Integers in decimal, without suffix outside `any`.
- Floats: the shortest decimal representation that parses back to the same value (bitwise), with `.0` appended if it has neither `.` nor exponent; exponent notation (`1.5e-7`, `1e21`) when the decimal exponent is lower than −6 or greater than 20; `inf`, `-inf`, `nan`. Implementations **MUST NOT** rely on the .NET Framework `"R"` format, which is not round-trip safe.
- Strings: single-line with the minimal escapes (`\"`, `\\`, `\n`, `\r`, `\t`, `\0`, `\u{…}` for other control characters); multi-line form when the string contains a line break and the value is written on its own line.
- `list<u8>`: Base64.
- Anchors: only on objects referenced more than once or exported, named `&1`, `&2` … in the order of first appearance; the object is written inline at its first appearance in field sequence order.
- Dependency aliases: the last segment of the dependency id if it is an identifier unique among aliases, otherwise `d1`, `d2` … by position.

9. Correspondence with the binary format

BINARY FEATURE	TEXT
Header fields, metadata, dependencies	directives (§5.1)
Schema modes Inline / Reference / None	`@schema { … }` / `@schema name version` / no `@schema`

BINARY FEATURE	TEXT
Exports	`@export`
Polymorphism tag	type prefix
`ref` object ids	anchors
`any` type descriptor	literal type, suffix or explicit type
Varints, fixed encoding, presence bitmap, string deduplication, packed bools	no equivalent (encoding details)
Compression	no equivalent; the whole file **MAY** be compressed externally, but it is then no longer indexable by search paths
Schema evolution (aliases, removed fields)	not applicable to text documents (§6.3); applies after conversion to binary

Text → binary conversion uses the writer options for `dedupMax`, compression and schema mode. Binary → text conversion of a file whose schema mode is Inline produces inline `@schema` blocks, rendered in the canonical schema language form.

10. Complete example

```
@sbdl 1
@id "level-01"
@version 3
@meta author = "Arthur"
@schema game 3
@dep tex = assets.textures 2
@dep items = game.items 1

Player {
  party: [
    Warrior { name: "Conan", rage: 12 }
    Mage { name: "Merlin", level: 7, mana: 300 }
  ]
  inventory: [
    { name: "Potion", count: 3 }
    { name: "Torch" }
  ]
  stats: { "str": 12.5, "dex": 8.0 }
  leader: &1 Warrior { name: "Conan", rage: 12 }
  icon: <tex>
  hash: 11473521213003792385
  experience: 1200
```

```
  extra: b64"AQID"
}

@export captain = *1
```

Appendix A. Complete grammar

```
(* Lexical *)
Letter       = "A".."Z" | "a".."z" | "_"
Digit        = "0".."9"
HexDigit     = Digit | "a".."f" | "A".."F"
BinDigit     = "0" | "1"
Ident        = Letter { Letter | Digit }
QName        = Ident { "." Ident }
Decimal      = Digit { [ "_" ] Digit }
Hex          = "0x" HexDigit { [ "_" ] HexDigit }
Binary       = "0b" BinDigit { [ "_" ] BinDigit }
IntSuffix    = "u8" | "u16" | "u32" | "u64" | "s8" | "s16" | "s32" | "s64"
FloatSuffix  = "f32" | "f64"
Integer      = [ "-" ] ( Decimal | Hex | Binary ) [ IntSuffix ]
Float        = [ "-" ] ( Decimal "." Decimal [ Exponent ] | Decimal Exponent | "inf" ) [
FloatSuffix ]
             | Decimal FloatSuffix
             | "nan" [ FloatSuffix ]
Exponent     = ( "e" | "E" ) [ "+" | "-" ] Decimal
String       = '"' { Char | Escape } '"'
MultiString  = '"""' LineBreak { Line } Indent '"""'
Base64       = 'b64"' { Base64Char | Whitespace | LineBreak } '"'
Escape       = "\" ( '"' | "\" | "n" | "r" | "t" | "0" | "u{" HexDigit { HexDigit } "}" )
Sep          = "," | LineBreak { LineBreak }

(* Types *)
IntType      = "u8" | "u16" | "u32" | "u64" | "s8" | "s16" | "s32" | "s64"
Type         = BaseType [ "?" ]
BaseType     = "bool" | IntType | "f32" | "f64" | "string" | "any"
             | "list" "<" Type ">" | "map" "<" Type "," Type ">"
             | "ref" "<" QName ">" | "extref" "<" QName ">" | QName

(* Schema language *)
SchemaFile   = "schema" QName Decimal { Import } { TypeDecl }
Import       = "import" QName Decimal [ Hex ]
TypeDecl     = EnumDecl | StructDecl
EnumDecl     = "enum" Ident ":" IntType "{" [ EnumValueDecl { Sep EnumValueDecl } [ "," ] ] "}"
EnumValueDecl = Ident [ "=" Integer ]
StructDecl   = [ "abstract" ] "struct" Ident [ ":" QName ] "{" [ FieldDecl { Sep FieldDecl } [
"," ] ] "}"
```

```
FieldDecl    = Ident ":" Type [ "=" Scalar ] { Annotation }
Annotation   = "@fixed" | "@alias" "(" Ident { "," Ident } ")"

(* Data document *)
Document     = FormatDirective { LineBreak } { Directive LineBreak } Braced { LineBreak } {
ExportDirective LineBreak }
FormatDirective = "@sbdl" Decimal
Directive    = "@id" String
             | "@version" Decimal
             | "@meta" ( Ident | String ) "=" String
             | "@schema" QName Decimal [ Hex ]
             | "@schema" "{" SchemaFile "}"
             | "@dep" Ident "=" ( QName | String ) Decimal
ExportDirective = "@export" ( Ident | String ) "=" ( Reference | [ Anchor ] Braced )

(* Values *)
Value        = "null" | Scalar | List | Braced | Anchor Braced | Reference | ExtRef |
TypedValue
Scalar       = "true" | "false" | Integer | Float | String | MultiString | Base64 | QName
List         = "[" [ Value { Sep Value } [ "," ] ] "]"
Braced       = [ QName ] "{" [ Entry { Sep Entry } [ "," ] ] "}"
Entry        = Key ":" Value
Key          = Ident | String | Integer | "true" | "false"
Anchor       = "&" ( Ident | Decimal )
Reference    = "*" ( Ident | Decimal )
ExtRef       = "<" Ident [ "#" ( Ident | String ) ] ">"
TypedValue   = Type Value
```

Appendix B. Conformance checklist for readers

A reader **MUST** reject: missing or misplaced `@sbdl`; unsupported format version; BOM or lone `CR`; invalid escape or control character in a string; malformed multi-line string indentation; invalid Base64; items on the same line without comma; duplicate directives, metadata keys, dependency aliases or ids, export names, anchors, struct fields or map keys; unknown or missing required fields; values not accepted for the expected type or out of range; unqualified enum value or unprefixed struct in `any`; missing mandatory type prefix; unresolved reference or undeclared dependency alias; anchor at a non-reference position; schema hash mismatch.